

Technical Documentation

RAG-Based Customer Support Assistant

1. Introduction

What is RAG?

RAG combines retrieval and generation to improve accuracy.

Why Needed?

- Reduces hallucination

- Uses real data

- Improves reliability

2. System Architecture

The system consists of:

- PDF ingestion pipeline

- Embedding and vector storage

- Retrieval layer

- LLM processing

- LangGraph workflow

- HITL module

3. Design Decisions

Chunk Size

800 tokens for balance between context and performance.

Embedding Strategy

Semantic embeddings for similarity search.

Retrieval

Top-K retrieval for relevant context.

Prompt Design

Ensures answers are grounded in context.

4. Workflow Explanation

LangGraph manages:

- State transitions

- Execution flow

- Conditional routing

5. Conditional Logic

Decision based on:

- Confidence score

- Context availability

6. HITL Implementation

Human intervention improves:

- Accuracy

- Reliability

Limitations:

- Increased response time

7. Challenges & Trade-offs

- Accuracy vs Speed

- Cost vs Performance

- Chunk size vs context

8. Testing Strategy

Test cases:

FAQ queries

Unknown queries

Edge cases

Metrics:

Accuracy

Latency

Escalation rate

9. Future Enhancements

Multi-document support

Chat memory

Deployment (cloud)

Feedback learning

Conclusion

This system demonstrates:

RAG architecture

Workflow-based AI system

Human-AI collaboration

It is a scalable solution for intelligent customer support automation.